

Lab 4 - Accelerated Convergence

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Problem 1

The following proves that this is infact a fixed point

$$\sqrt{2} = \sqrt{\frac{10 - \sqrt{2}^2}{4}}$$

$$\sqrt{2} = \sqrt{\frac{10 - 2}{4}}$$

$$\sqrt{2} = \sqrt{\frac{8}{4}}$$

$$\sqrt{2} = \sqrt{2}$$

$$0 = 0$$

The code yields the following result:

n	xn	E	En1	En2	En3
0	1.5	0	0	0	0
1	1.3919	0.10806	0	0	0
2	1.4197	0.027786	0.25714	2.3796	22.021
3	1.4128	0.0068953	0.24816	8.931	321.42
4	1.4146	0.001727	0.25046	36.323	5267.8
5	1.4141	0.00043154	0.24989	144.7	83786
6	1.4142	0.0001079	0.25003	579.38	1.3426e+06
7	1.4142	2.6974e-05	0.24999	2316.9	2.1473e+07
8	1.4142	6.7435e-06	0.25	9268.3	3.436e+08
9	1.4142	1.6859e-06	0.25	37073	5.4975e+09
10	1.4142	4.2147e-07	0.25	1.4829e+05	8.7961e+10
11	1.4142	1.0537e-07	0.25	5.9316e+05	1.4074e+12
12	1.4142	2.6342e-08	0.25	2.3727e+06	2.2518e+13
13	1.4142	6.5854e-09	0.25	9.4906e+06	3.6029e+14
14	1.4142	1.6464e-09	0.25	3.7963e+07	5.7646e+15
15	1.4142	4.1159e-10	0.25	1.5185e+08	9.2234e+16
16	1.4142	1.029e-10	0.25	6.074e+08	1.4757e+18
17	1.4142	2.5724e-11	0.25	2.4296e+09	2.3612e+19
18	1.4142	6.4311e-12	0.25	9.7184e+09	3.7779e+20
19	1.4142	1.6078e-12	0.25001	3.8875e+10	6.0449e+21

Problem 2

Code:

```
xf = MyFixedPoint(1.5,10^(-15),20,@(x) sqrt((10-x^2)/4))  
MyAitken(xf,size(xf,1))
```

```

function [result] = MyAitken(xf,n)
    xn = 0;
    x1 = 0;
    xs = [0; 0];
    i = 0;
    N = [0; 1];
    E= [0; 0; 0];
    En1 = [0; 0; 0; 0];
    En2 = [0; 0; 0; 0];
    En3 = [0; 0; 0; 0];
    while i+2 < n
        l = (xf(3+i,1)-xf(2+i,1))/(xf(2+i,1)-xf(1+i,1));
        xn = xf(3+i,1)+((1/(1-l))*(xf(3+i,1)-xf(2+i,1)));
        xs = [xs; xn];
        i = i + 1;
        if i >= 2
            E = [E; abs(x1-xn)];
        end
        if i >= 3
            En1 = [En1; (abs(x1-xn))/(en)];
            En2 = [En2; (abs(x1-xn))/(en^2)];
            En3 = [En3; (abs(x1-xn))/(en^3)];
        end
        en = abs(x1-xn);
        x1 = xn;
        N = [N; i+1];
    end
    xn = xs;
    T = table(N,xf,xn,E,En1,En2,En3);
    result = T;
end

function [result] = MyFixedPoint(x1,tol,Nmax,g)
arguments
    x1 double
    tol double
    Nmax double
    g function_handle
end
    xs = [x1];
    ys = [g(x1)];
    xn = x1;
    E = [0];
    En1 = [0; 0];
    En2 = [0; 0];
    En3 = [0; 0];
    n = [0];
    i = 1;
    while abs(g(xn)-xn) > tol && i < Nmax

```

```

n = [n; i];
xn = g(x1);
xs = [xs; xn];
ys = [ys; g(xn)];
E = [E; abs(x1-xn)];
if i > 1
    En1 = [En1; (abs(x1-xn))/(en)];
    En2 = [En2; (abs(x1-xn))/(en^2)];
    En3 = [En3; (abs(x1-xn))/(en^3)];
end
en = abs(x1-xn);
x1 = xn;
i = i + 1;
end
xn = xs;
yn = ys;
T = table(n,xn,E,En1,En2,En3);
%result = T;
result = xs
end

```

Output:

20×7 table

N	xf	xn	E	En1	En2	En3
0	1.5	0	0	0	0	0
1	1.3919	0	0	0	0	0
2	1.4197	1.414	0	0	0	0
3	1.4128	1.4142	0.00015904	0	0	0
4	1.4146	1.4142	1.0165e-05	0.063911	401.85	2.5267e+06
5	1.4141	1.4142	6.3145e-07	0.062123	6111.7	6.0127e+08
6	1.4142	1.4142	3.9524e-08	0.062593	99125	1.5698e+11
7	1.4142	1.4142	2.4694e-09	0.062477	1.5807e+06	3.9993e+13
8	1.4142	1.4142	1.5435e-10	0.062506	2.5313e+07	1.0251e+16
9	1.4142	1.4142	9.6465e-12	0.062498	4.0491e+08	2.6234e+18
10	1.4142	1.4142	6.0307e-13	0.062517	6.4808e+09	6.7183e+20
11	1.4142	1.4142	3.7526e-14	0.062224	1.0318e+11	1.7109e+23
12	1.4142	1.4142	2.4425e-15	0.065089	1.7345e+12	4.6222e+25
13	1.4142	1.4142	2.2204e-16	0.090909	3.722e+13	1.5238e+28
14	1.4142	1.4142	0	0	0	0
15	1.4142	1.4142	0	NaN	NaN	NaN
16	1.4142	1.4142	2.2204e-16	Inf	Inf	Inf
17	1.4142	1.4142	0	0	0	0
18	1.4142	1.4142	0	NaN	NaN	NaN
19	1.4142	1.4142	0	NaN	NaN	NaN

Problem 3

Checking fixed point:

a)

$$\sqrt{2} = \frac{1}{2}(\sqrt{2} + \frac{2}{\sqrt{2}})$$

$$\sqrt{2} = \frac{1}{2}(\sqrt{2} + \sqrt{2})$$

$$\sqrt{2} = \sqrt{2}$$

$$0 = 0$$

b)

$$\sqrt{2} = \frac{2}{8\sqrt{2}} + \sqrt{2} - \frac{1}{4\sqrt{2}}$$

$$\sqrt{2} = \frac{1}{4\sqrt{2}} + \sqrt{2} - \frac{1}{4\sqrt{2}}$$

$$\sqrt{2} = \sqrt{2}$$

$$0 = 0$$

c)

$$\sqrt{2} = \frac{12 + 24 - 4}{16\sqrt{2}}$$

$$\sqrt{2} = \sqrt{2} \frac{12 + 24 - 4}{32}$$

$$\sqrt{2} = \sqrt{2}$$

$$0 = 0$$

Outputs:

a)

5x7 table

N	xf	xn	E	En1	En2	En3
0	1.5	0	0	0	0	0
1	1.4167	0	0	0	0	0
2	1.4142	1.4141	0	0	0	0
3	1.4142	1.4142	7.2146e-05	0	0	0
4	1.4142	1.4142	1.8405e-09	2.551e-05	0.35359	4901

Clearly not worth it, Newton's method converges faster than Aitken's can help it speed it

b)

20x7 table

N	xf	xn	E	En1	En2	En3
0	1.5	0	0	0	0	0
1	1.5221	0	0	0	0	0
2	1.5501	1.4173	0	0	0	0
3	1.5857	1.419	0.0017093	0	0	0
4	1.6312	1.4216	0.0026593	1.5558	910.22	5.3252e+05
5	1.6896	1.4258	0.0041346	1.5548	584.66	2.1985e+05
6	1.7651	1.4322	0.0064245	1.5538	375.81	90894
7	1.8637	1.4422	0.0099783	1.5532	241.76	37630
8	1.994	1.4577	0.015497	1.5531	155.65	15599
9	2.1686	1.4818	0.024087	1.5542	100.29	6471.4
10	2.4075	1.5193	0.037514	1.5575	64.66	2684.5
11	2.743	1.578	0.058678	1.5642	41.695	1111.5
12	3.2313	1.6705	0.092515	1.5767	26.869	457.91
13	3.9774	1.8184	0.14789	1.5986	17.279	186.77
14	5.1989	2.0603	0.24194	1.6359	11.062	74.795
15	7.4111	2.4714	0.41107	1.699	7.0225	29.026
16	12.089	3.2141	0.7427	1.8068	4.3954	10.693
17	24.83	4.697	1.483	1.9967	2.6884	3.6198
18	79.145	8.1846	3.4876	2.3518	1.5859	1.0694
19	632.63	18.919	10.735	3.078	0.88256	0.25306

Method diverges, so Aitken can't help us out here very much

c)

4x7 table

N	xf	xn	E	En1	En2	En3
0	1.5	0	0	0	0	0
1	1.4144	0	0	0	0	0
2	1.4142	1.4142	0	0	0	0
3	1.4142	1.4142	2.2365e-07	0	0	0

once again, the fixed-point iteration converges so quickly that there's no point in Aitken's